PK Prozessdatenverarbeitung und eingebettete Systeme

Anmeldung via e-Learning
(ohne Passwort)

Gruppeneinteilung: Bitte tragen Sie sich bis 24.03.2026 (23:59 Uhr) in eine der Gruppen ein.

Voice Control

Introduction

Parallel classes

- German:
  - Thursday 10:00 – 13:15 (4 SWS)
  - Room 5.1.06
  - BET: Prozessdatenverarbeitung und eingebettete Systeme
  - BET: Sprachsteuerung (SPO 2025)
  - BET: Sprachsteuerung (ältere SPO kann nicht mehr separat angeboten werden)
- English:
  - Thursday 14:15 – 15:00 (1 SWS)
  - Room 1.E.33
  - IRO: Speech recognition and speech synthesis
  - BRO: Spracherkennung und Sprachsynthese

- Identical content, identical exam

Requirements

- Hardware: Bring your own device – Notebook
- Software:
  - Python, e.g. 3.11 or higher
  - Jupyter Notebooks

Goals and motivation

- Python programming,
- Pair programming and test driven development
- Audio analysis and signal processing,
- Denoising,
- Machine Learning

- 12 Weeks
  - Week01_PythonTutorial

  ...
  - Week12_MachineLearning_SpeechToText_TextToSpeech
- Each week corresponds to 1 Jupyter Notebook
  - explains a single topic
  - Gives you a programming exercise
  - Gives you a set of tasks for exam preparation

Tasks for German course

Asynchronous tasks part A (2 hours):

- Download the files of the week
- For last week: Compare own solutions with reference
- Read and understand the weekly Jupyter Notebook
- Read the weekly tasks for exam preparation
- Prepare and publish in FAQ one question regarding the content of
  - Jupyter Notebook
  - Exam preparation

Synchronous tasks (4 hours):

- Discuss questions in FAQ
- Discuss Jupyter Notebook
- Solve the programming exercise
- Discuss first exam preparation tasks

Asynchronous tasks part B (2 hours):

- Solve exam preparation tasks

Link to eLearning

Tasks for English course

Asynchronous tasks part A (6 hours):

- Download the files of the week
- For last week: Compare own solutions with reference
- Read and understand the weekly Jupyter Notebook
- Read the weekly tasks for exam preparation
- Prepare and publish in FAQ one question regarding the content of
  - Jupyter Notebook
  - Exam preparation
- Solve the programming exercise
- Solve first exam preparation tasks

Synchronous tasks (1 hours):

- Discuss questions in FAQ
- Discuss open points for Jupyter Notebook
- Discuss open points for exam preparation tasks

Asynchronous tasks part B (1 hours):

- Solve exam preparation tasks

Link to eLearning

Tasks (short version)

At home:

- Every Monday: Download the files from the cloud.
- Work with the current Jupyter Notebook.
- Publish two questions in FAQ:
  - One regarding the Jupyter Notebook and
  - One regarding the exam preparations.

Link to cloud files

Time schedule

- 1 week: Python tutorial
- 5 weeks: theory part A
- 1 week: mock exam
- 5 weeks: theory part B
- Remaining weeks: exam preparation

Time schedule

- Each Monday, the cloud is already updated
  - download newest files
  - Read the tasks of the week (Focus: work with the Jupyter Notebook)
  - Do the tasks of the week☺

- Week 1: 16.03. – 20.03.
- Week 2: Acoustic Conference and Easter: no lessons, but FAQ and self learning
- On Sunday, 05.04.,
  - the software should run and
  - Week 1 and week 2 should be finished by you.
- Week 3: 06.04. – 10.04.
- Week 4: 13.04. – 17.04.
- …

Support for the students

- Weekly support:
  - use the time in the class room for questions
  - Every weekend, the reference solutions are published
  - Normal consultation hours: please take a look at my website (google->“spiertz thws“)
- Midterm support: mock exam
- Asynchronous feedback by FAQ forum (moderated by me)
- Immediate feedback for programming exercises by unit tests

Grade improvement

- Special task in april and may:
- Successfull participating means: if you pass the exam, your grade will be improved by one step, e.g. 3.7 -> 3.3
- I need four voice recordings (2 female, 2 male) under different background noise situation.
- The recording time will be in the range of 5-10 minutes per person.
- These recordings will be **published** anonymously.
- I will also participate.

- The scientific topic: the lombard effect

Summary

- Until 05.04.
  - Install recent Python version
  - Install Jupyter Notebook
  - (Visual Studio Code and Github Copilot are optionally recommended)
- After 05.04. start your weekly schedule:
  - Monday: download recent files
  - Until Thursday:
    - Work with the material
    - Publish two questions in FAQ (one regarding the Jupyter Notebook, one regarding the exam preparations)

Thanks a lot for your attention...

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